EASY · ADVANCED MATH · ANSWER KEY

Equivalent expressions

03

10 answers · Rationale-rich · Teach from doc

Q1

D

D. $16x^2 + 9x$

Choice D is correct. In the given expression, the first two terms, $9x^2$ and $7x^2$, are like terms. Combining these like terms yields $9x^2 + 7x^2$, or $16x^2$. It follows that the expression $9x^2 + 7x^2 + 9x$ is equivalent to $16x^2 + 9x$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q2

D

D. $2a^3 + 6a^2$

Choice D is correct. Expanding the given expression using the distributive property yields $2a^2(a) + 2a^2(3)$. Combining like terms yields $2a^2(a^1) + (2 \times 3)(a^2)$, or $2a^{2+1} + 6a^2$, which is equivalent to $2a^3 + 6a^2$.

Choices A and B are incorrect and may result from incorrectly combining like terms. Choice C is incorrect and may result from distributing $2a^2$ only to a, and not to 3, in the given expression.

Q3

B

B. $6x^3$

Choice B is correct. The given expression can be rewritten as $11x^3 + -5x^3$. Since the two terms of this expression are both constant multiples of x^3, they are like terms and can, therefore, be combined through addition. Adding like terms in the expression $11x^3 + -5x^3$ yields $6x^3$.

Choice A is incorrect. This is equivalent to $11x^3 + 5x^3$, not $11x^3 - 5x^3$.

Choice C is incorrect. This is equivalent to $11x^6 - 5x^6$, not $11x^3 - 5x^3$.

Choice D is incorrect. This is equivalent to $11x^6 + 5x^6$, not $11x^3 - 5x^3$.

Q4

B

B. $2xy8x^2y + 7$

Choice B is correct. Since $2xy$ is a common factor of each term in the given expression, the expression can be rewritten as $2xy8x^2y + 7$.

Choice A is incorrect. This expression is equivalent to $16x^2y^2 + 14xy$.

Choice C is incorrect. This expression is equivalent to $28x^3y^2 + 14xy$.

Choice D is incorrect. This expression is equivalent to $112x^3y^2 + 14xy$.

Q5

C

C. $x23x^2 + 2x + 9$

Choice C is correct. Since x is a common factor of each term in the given expression, the given expression can be rewritten as $x23x^2 + 2x + 9$.

Choice A is incorrect. This expression is equivalent to $23x^3 + 46x^2 + 207x$.

Choice B is incorrect. This expression is equivalent to $207x^4 + 18x^3 + 9x$.

Choice D is incorrect. This expression is equivalent to $34x^3 + 34x^2 + 34x$.

Q6

D

D. $55x^2$

Choice D is correct. The given expression shows addition of two like terms. Therefore, the given expression is equivalent to $50 + 5x^2$, or $55x^2$.

Choice A is incorrect. This expression is equivalent to $505x^2$, not $50 + 5x^2$.

Choice B is incorrect. This expression is equivalent to $\frac{50}{5}x^2$, not $50 + 5x^2$.

Choice C is incorrect. This expression is equivalent to $50 - 5x^2$, not $50 + 5x^2$.

Q7

A

A. $4x^2 + 24$

Choice A is correct. The expression $4\left(x^2 + 6\right)$ can be rewritten as $4x^2 + 4\left(6\right)$, which is equivalent to $4x^2 + 24$.

Choice B is incorrect. This expression is equivalent to $4x^2 + \frac{5}{2}$, not $4x^2 + 6$.

Choice C is incorrect. This expression is equivalent to $4x^2 + \frac{3}{2}$, not $4x^2 + 6$.

Choice D is incorrect. This expression is equivalent to $4x^2 - \frac{1}{2}$, not $4x^2 + 6$.

Q8

D

D. $r^3+6r^2+8r+19$

Choice D is correct. Grouping like terms, the given expressions can be rewritten as $r^3+(5r^2+r^2)+8r+(7+12)$. This can be rewritten as $r^3+6r^2+8r+19$.

Choice A is incorrect and may result from adding the two sets of unlike terms, r^3 and r^2 as well as $5r^2$ and $8r$, and then adding the respective exponents. Choice B is incorrect and may result from adding the unlike terms r^3 and r^2 as if they were r^3 and r^3 and adding the unlike terms $5r^2$ and $8r$ as if they were $5r^2$ and $8r^2$. Choice C is incorrect and may result from errors when combining like terms.

Q9

A

A. $5x^2-5x$

Choice A is correct. Since (x^2-x) is a common term in the original expression, like terms can be added: $2(x^2-x)+3(x^2-x)=5(x^2-x)$. Distributing the constant term 5 yields $5x^2-5x$.

Choice B is incorrect and may result from not distributing the negative signs in the expressions within the parentheses. Choice C is incorrect and may result from not distributing the negative signs in the expressions within the parentheses and from incorrectly eliminating the x^2-term. Choice D is incorrect and may result from incorrectly eliminating the x-term.

Q10 B

B. $x^3 5x^2 - 6x + 8$

Choice B is correct. Since x^3 is a common factor of each term in the given expression, the expression can be rewritten as $x^3 5x^2 - 6x + 8$.

Choice A is incorrect. This expression is equivalent to $5x^5 - 6x^4$.

Choice C is incorrect. This expression is equivalent to $40x^5 - 48x^4 + 8x^3$.

Choice D is incorrect. This expression is equivalent to $-36x^9 + 48x^8 + 6x^5$.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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